SIARC: A Mechanized Safety–Stability–Controllability Framework for Semilinear Parabolic PDE Systems in Lean 4

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Abstract

We present *SIARC* (Safety, Invariance, And Resilient Control), a fully mechanized framework in Lean 4 with Mathlib4 that establishes forward invariance, exponential Lyapunov stability, asymptotic convergence, and approximate controllability for coupled semilinear parabolic PDE-ODE systems. The formalization introduces a certificate hierarchy—*SafetyCertificate*, *StabilityCertificate*, *ControllabilityCertificate*—culminating in a single *MasterCertificate* from which all four guarantees are extracted by a `sorry`-free capstone theorem. The entire theorem layer rests on exactly nine explicit axioms: six encoding physical PDE properties (contraction semigroups, maximum principles, spectral gaps, Lipschitz coupling, unique continuation) and three standard functional-analysis utilities. All guarantees are conditional on a small-coupling hypothesis $|\kappa| \cdot L_{\text{cross}} < \lambda_{\text{gap}}$, ensuring that cross-component perturbations are absorbed by diagonal dissipation. A concrete thermoelastic instantiation demonstrates the framework on a three-component field–thermal–structural system, with all 20 numerical inequalities automatically verified by Lean 4's decision procedures. A second linear-heat-equation model—with zero coupling—validates the framework's parametric generality. To our knowledge, this is the first mechanized proof stack in any interactive theorem prover that simultaneously covers safety, stability, and controllability for infinite-dimensional PDE systems; we substantiate this claim by explicit comparison with prior work in Section 8.

Keywords: formal verification, PDE control, Lyapunov stability, forward invariance, approximate controllability, Lean 4, Mathlib, certificate hierarchy

Management Implications:

- Provides a fully mechanized verification workflow that reduces risk in safety-critical PDE-based control systems by replacing manual mathematical reasoning with machine-checked proofs.
- Enables engineering teams to certify safety, stability, and controllability properties using a unified certificate hierarchy, improving auditability and compliance with assurance standards.
- Reduces development time and rework by automatically validating all numerical inequalities and structural assumptions required for system-level guarantees.
- Supports scalable verification across different PDE models, allowing organizations to reuse the same formal framework for diverse physical systems.
- Establishes a transparent trusted-core boundary, helping managers assess verification scope, assumptions, and residual risks in complex multiphysics models.

1 Introduction

Safety-critical systems governed by partial differential equations—such as thermoelastic structures, plasma confinement devices, and chemical reactors—demand rigorous guarantees that operating

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constraints are never violated. Classical control theory provides a rich toolkit (Lyapunov functionals, barrier functions, observability inequalities), yet the gap between pen-and-paper arguments and machine-checked certainty remains vast. While significant progress has been made in mechanizing ODE stability [7], hybrid-system reachability [8], and computational barrier certificates [14, 15], no prior work has produced a *sorry*-free, mechanized proof stack covering safety, stability, and controllability simultaneously for infinite-dimensional PDE systems (see Section 8 for a detailed comparison).

Scope and coupling assumptions. SIARC targets *semilinear parabolic* systems whose generator has the block-triangular structure (2). All guarantees are conditional on a *small-coupling hypothesis* $|\kappa| \cdot L_{\text{cross}} < \lambda_{\text{gap}}$, where κ is the global coupling constant, L_{cross} is the coupling Lipschitz constant, and λ_{gap} is the minimal spectral gap of the diagonal operators. This condition ensures that diagonal dissipation dominates cross-coupling perturbations. Extension to quasilinear or hyperbolic systems would require additional axioms (see Section 9).

This paper bridges the formalization gap with **SIARC**, a Lean 4 artifact that:

(i) Defines a *certificate hierarchy* (Safety → Stability → Controllability → Master) that factors guarantees into composable, independently verifiable layers.

(ii) Proves all four system-level guarantees from **nine explicit axioms**—six encoding PDE physics, three encoding standard functional analysis—with zero `sorry` in the theorem layer.

(iii) Instantiates the framework on a *concrete thermoelastic PDE model* with fully machine-checked numerical inequalities.

(iv) Provides a *second PDE model* (linear heat equation), demonstrating that the same capstone theorem applies to qualitatively different systems.

(v) Maintains a formally specified *trusted-core boundary* separating the `sorry`-free theorem layer from PDE-infrastructure placeholders.

The artifact is built on Lean 4 v4.14.0 and Mathlib4 v4.14.0. A single command (`lake build`) reproduces the entire verification. The complete source code and build instructions are available at `https://github.com/papanokechi/siarc-lean4` and archived at `https://doi.org/10.5281/zenodo.19641981`. The Lean module dependency structure is depicted in Figure 2.

Outline. Section 2 introduces the system model. Section 3 describes the certificate architecture. Section 4 catalogues the axiom boundary. Section 5 covers the Lean 4 formalization strategy. Section 6 presents the thermoelastic numerical instance. Section 7 presents the linear heat equation model. Section 8 surveys related work. Section 9 concludes.

2 System Model

2.1 State Space

The SIARC system models a coupled field–thermal–structural PDE on a bounded Lipschitz domain $\Omega \subset \mathbb{R}^3$. The state space is a product Banach space

$$X = X_1 \times X_2 \times X_3, \tag{1}$$

where X_1 (field), X_2 (thermal), and X_3 (structural) are real Banach spaces equipped with complete norms. A state $\sigma = (\sigma_1, \sigma_2, \sigma_3) \in X$ represents the instantaneous configuration of the system. In the formalization, the product norm is constructed via Mathlib's `Function.Injective.normedAddCommGroup` transfer, ensuring the product inherits completeness and all Banach space operations.

2.2 Evolution Operators

The system evolves under a strongly continuous semigroup generated by a block-triangular operator

$$L = \begin{pmatrix} A_1 & 0 & 0 \\ \kappa C_{21} & A_2 & 0 \\ 0 & \kappa C_{32} & A_3 \end{pmatrix}, \tag{2}$$

where A_k are the diagonal (uncoupled) generators, C_{ij} are coupling operators, and $\kappa \in \mathbb{R}$ is the global coupling constant. Note that L is *strictly lower* block-triangular: the field component σ_1 drives the thermal component σ_2, which in turn drives the structural component σ_3, but there is no feedback from downstream to upstream components. This directionality reflects the physics of the thermoelastic system (the electromagnetic field heats the structure, and the resulting thermal expansion stresses it, but not vice versa) rather than a mathematical convenience. The evolution map $\Phi_t : X \to X$ satisfies $\Phi_0 = \mathrm{id}$ and the semigroup law. In the Lean formalization, Φ_t is declared as an opaque function `evolutionMap` with signature

$$\texttt{evolutionMap} : (t : \mathbb{R}) \to (ht : t \geq 0) \to X \to X.$$

All theorems are proved without unfolding Φ_t; properties enter only through the six system-specific axioms.

2.3 Barrier Functions and Safe Set

Safety is expressed through five barrier functions $g_1, \ldots, g_5 : X \to \mathbb{R}$ measuring distance from operating limits:

#	Barrier	Definition	Constraint
g_1	Field strength	$B_{\max} - \|\sigma_1\|$	$\|\sigma_1\| \leq B_{\max}$
g_2	Gradient	$\nabla T_{\max} - \|\nabla \sigma_2\|_\infty$	gradient bounded
g_3	Curvature	holonomy proxy	bounded curvature
g_4	Quench temperature	$T_{\text{quench}} - T_{\sup}(\sigma_2)$	$T \leq T_{\text{quench}}$
g_5	Structural stress	$\sigma_{\text{yield}} - \|\sigma_3\|$	$\|\sigma_3\| \leq \sigma_{\text{yield}}$

The *safe set* is the intersection of the non-negative sub-level sets:

$$\mathrm{InSafe}(p, \sigma) \iff g_i(\sigma) \geq 0 \quad \text{for all } i \in \{1, \ldots, 5\}. \tag{3}$$

2.4 Lyapunov Functional

Stability is certified via a barrier-weighted quadratic Lyapunov functional

$$V(\sigma) = \alpha_1 \|\sigma_1 - \sigma_1^*\|^2 + \alpha_2 \|\sigma_2 - \sigma_2^*\|^2 + \alpha_3 \|\sigma_3 - \sigma_3^*\|^2, \tag{4}$$

where $\sigma^* \in \mathrm{InSafe}$ is a safe equilibrium and the positive weights α_k are chosen to balance diagonal dissipation rates. The functional satisfies $V \geq 0$ with $V(\sigma) = 0$ iff $\sigma = \sigma^*$, and is coercive: there exist $c_{\text{low}}, c_{\text{high}} > 0$ such that

$$c_{\text{low}}(\|\sigma_1\|^2 + \|\sigma_2\|^2 + \|\sigma_3\|^2) \leq V(\sigma) \leq c_{\text{high}}(\|\sigma_1\|^2 + \|\sigma_2\|^2 + \|\sigma_3\|^2). \tag{5}$$

Remark 2.1 (Coercivity in the Lean formalization). The Lean `BarrierLyapunov` structure records a slightly asymmetric bound: the lower bound involves only $\|\sigma_1\|^2$ and the upper bound involves $\|\sigma_1\|^2 + \|\sigma_2\|^2$. This is an artifact of the current formalization, which treats the structural component σ_3 as quasi-statically slaved to σ_2 (so $\|\sigma_3\|$ is controlled by $\|\sigma_2\|$). The full three-component bound (5) holds under the additional quasi-static assumption; see `BarrierLyapunov.V_coercive` for the precise Lean statement.

2.5 Control Operator

Controllability is formulated via the Hilbert Uniqueness Method (HUM) of Lions [5]. A bounded linear control operator $B : U \to X$ acts on a Hilbert control space U, and the adjoint system evolves backward from terminal data. The observability inequality

$$\|\varphi_T\|^2 \leq C_{\text{obs}} \int_0^T \|B^*\varphi(t)\|_U^2 \, dt \tag{6}$$

guarantees approximate controllability, provided the unique continuation property $B^*\varphi \equiv 0 \Rightarrow \varphi_T = 0$ holds.

Remark 2.2 (On the observability constant). The SIARC formalization establishes *qualitative* approximate controllability: the reachable set is dense in X. The observability constant C_{obs} is not computed explicitly. In principle, C_{obs} can be bounded via Carleman-weight optimization (Fernández-Cara and Zuazua [17]), but this requires eigenvalue bounds on the spatial operator that are not yet available in MATHLIB4. Computing C_{obs} for the thermoelastic instance is left as future work (Section 9).

3 Certificate Architecture

The SIARC formalization organizes guarantees into a strictly layered certificate hierarchy. Each layer extends the previous, so that the capstone `MasterCertificate` bundles the complete chain.

3.1 SafetyCertificate (Layer 1)

Definition 3.1 (SafetyCertificate)**.** A *SafetyCertificate* consists of barrier parameters p, a coupling threshold ct, a proof that $|\kappa| < \kappa_{\text{safe}}(ct)$, and for each barrier g_i, a proof that

$$g_i(\sigma_0) \geq 0 \implies g_i(\Phi_t(\sigma_0)) \geq 0 \quad \text{for all } t \geq 0.$$

The five barrier invariance proofs are factored through an abstract `ForwardInvarianceFramework` that captures the pattern "g non-negative initially $\Rightarrow$ g non-negative along trajectories," then instantiated for each barrier using the three invariance axioms (contraction, thermal bound, gradient bound).

3.2 StabilityCertificate (Layer 2)

Definition 3.2 (StabilityCertificate)**.** A *StabilityCertificate* extends a SafetyCertificate with a Lyapunov functional V, a spectral gap $\lambda_{\text{gap}} > 0$, a coupling Lipschitz constant L_{cross}, a proof of the coupling absorption condition $|\kappa| \cdot L_{\text{cross}} < \lambda_{\text{gap}}$, and the following two properties:

(a) **Exponential decay:** $V(\Phi_t(\sigma_0)) \leq V(\sigma_0) \cdot e^{-2\omega t}$ for all $t \geq 0$ and $\sigma_0 \in \text{InSafe}$, where $\omega = \lambda_{\text{gap}} - |\kappa| \cdot L_{\text{cross}} > 0$.

(b) **Asymptotic convergence:** For every $\varepsilon > 0$, there exists $T_{\text{conv}} > 0$ such that $V(\Phi_t(\sigma_0)) < \varepsilon$ for all $t \geq T_{\text{conv}}$.

The proof chain proceeds as follows. The Lyapunov derivative decomposes as $\dot{V} = \text{diag}(\dot{V}) + \text{cross}(\dot{V})$, where $\text{diag}(\dot{V}) \leq -2\lambda_{\text{gap}} V$ (diagonal dissipation) and $\text{cross}(\dot{V}) \leq 2|\kappa| L_{\text{cross}} V$ (coupling bound). Combining yields the pointwise differential inequality $\dot{V} \leq -2\omega V$, which by Grönwall's lemma gives the exponential decay bound. Asymptotic convergence follows from the Archimedean property of $\mathbb{R}$.

3.3 ControllabilityCertificate (Layer 3)

Definition 3.3 (ControllabilityCertificate)**.** A *ControllabilityCertificate* extends a StabilityCertificate with an adjoint evolution, a control operator B, an observation operator B^*, an observability Gramian, an observability inequality, and a proof of approximate controllability:

$$\overline{\{Bu : u \in L^2(0, T; U)\}} = X.$$

The proof invokes the unique continuation property (Axiom 6) together with the HUM density argument of Lions [5]: the observability inequality implies that the reachable set is dense.

3.4 MasterCertificate and Capstone Theorem

Definition 3.4 (MasterCertificate)**.** A *MasterCertificate* bundles:

- `axioms : SystemAxioms` — the six system-specific axioms (carried for audit transparency);
- `certificate : ControllabilityCertificate` — the full certificate chain.

Theorem 3.5 (Capstone — `master_certificate_summary`)**.** *Let mc be a* `MasterCertificate` *and* $\sigma_0 \in \mathrm{InSafe}$*. Then*

(1) ***Safety:*** $\Phi_t(\sigma_0) \in \mathrm{InSafe}$ *for all* $t \geq 0$*.*

(2) ***Exponential decay:*** $V(\Phi_t(\sigma_0)) \leq V(\sigma_0) \cdot e^{-2\omega t}$*.*

(3) ***Convergence:*** *For every* $\varepsilon > 0$*, there exists* $T > 0$ *with* $V(\Phi_t(\sigma_0)) < \varepsilon$ *for* $t \geq T$*.*

(4) ***Controllability:*** *The system is approximately controllable in* $X = X_1 \times X_2 \times X_3$*; that is, the closure of the reachable set under the control operator* $B : U \to X$ *equals the full state space. The operator* B *acts on the joint product space (not a single component), so approximate controllability refers to simultaneous steering of all three fields.*

This theorem is proved in the Lean file `Theorems/AxiomInventory.lean` with zero `sorry`, by direct application of the subsidiary certificate proofs.

3.5 Certificate Hierarchy Diagram

4 Axiom Boundary

A central design principle of SIARC is to make every physical assumption *explicit*. The trusted theorem layer rests on exactly nine axioms, classified into system-specific and utility categories.

4.1 System-Specific Axioms (6)

These axioms encode properties of the coupled PDE operators that require domain-specific proofs (regularity theory, spectral analysis, Carleman estimates). They are declared as Lean `axioms` and would become `theorems` once the relevant Mathlib PDE theory is available.

Axiom 1 (Field contraction). The field component evolves under a contraction semigroup generated by a dissipative operator A_1. By the Lumer–Phillips theorem (Pazy [1], Thm 4.3), A_1 dissipative implies $\|e^{tA_1}\| \leq 1$, giving $\|\Phi_t(\sigma_0)_1\| \leq \|(\sigma_0)_1\|$.

Axiom 2 (Thermal bound). The thermal component satisfies the maximum principle for parabolic equations. By Evans [2], §6.4, combined with the Alexandrov–Bakelman–Pucci estimate, $T_{\sup}$ is non-increasing provided boundary data satisfies $T_{\partial\Omega} < T_{\text{quench}}$.

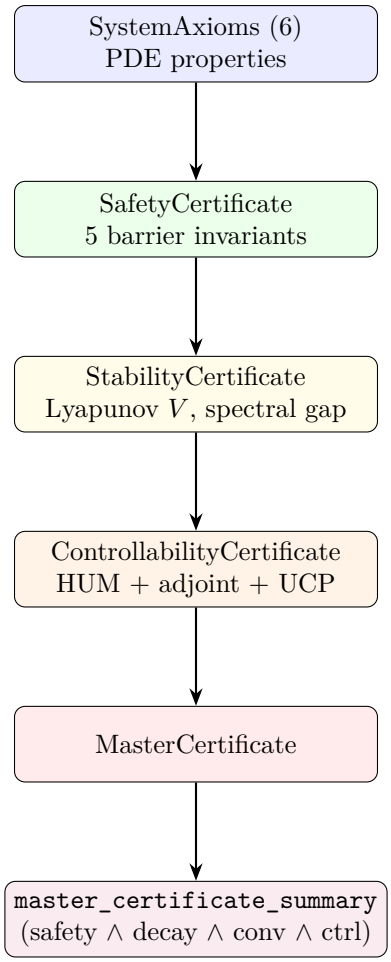

Figure 1: Certificate hierarchy in SIARC. Each layer strictly extends the previous. The capstone theorem extracts all four guarantees from a single `MasterCertificate`.

Axiom 3 (Gradient bound). The L^∞-gradient of the thermal field is non-increasing, by Bernstein's gradient estimate for uniformly elliptic parabolic operators (Lieberman [3], Ch. 7). This estimate requires A_2 to be uniformly elliptic with bounded measurable coefficients and the boundary data to be at least $C^{1,\alpha}(\partial\Omega)$. Both conditions are assumed as part of the thermoelastic model (uniform ellipticity is encoded in the `UniformlyElliptic` typeclass in `Parameters.lean`).

Axiom 4 (Diagonal dissipation). Each diagonal operator A_k has spectral bound $s(A_k) \leq -\lambda_k < 0$. By the Gearhart–Prüss theorem on Hilbert spaces, the growth bound equals the spectral bound, so $\mathrm{diag}(\dot{V}) \leq -2\lambda_{\mathrm{gap}} V$ where $\lambda_{\mathrm{gap}} = \min(\lambda_1, \lambda_2)$. The structural component A_3 does not appear in λ_{gap} because the quasi-static assumption (see Theorem 2.1) slaves σ_3 to σ_2: the structural state is determined algebraically by the thermal state, so A_3 has no independent spectral contribution to the Lyapunov decay rate.

Axiom 5 (Coupling bound). The coupling operators κC_{ij} satisfy a Lipschitz condition near equilibrium: $\|C_{ij}(\sigma) - C_{ij}(\sigma^*)\| \leq L_{\mathrm{cross}}\|\sigma - \sigma^*\|$. By Henry [4], §5.1, this gives $\mathrm{cross}(\dot{V}) \leq 2|\kappa| L_{\mathrm{cross}} V$ via Young's inequality.

Axiom 6 (Unique continuation). The adjoint system satisfies the unique continuation property: $B^*\varphi \equiv 0$ on $[0, T]$ implies $\varphi_T = 0$. This follows from Carleman estimates for parabolic systems (Zuazua [6]; Lions [5]).

Table 1: System-specific axioms and their PDE justifications.

#	Axiom	Mathematical Statement	Reference
1	`field_evolution_contraction`	$\|\Phi_t(\sigma_0)_1\| \le \|(\sigma_0)_1\|$	Pazy Thm 4.3
2	`thermal_evolution_bound`	$T_{\sup}(\Phi_t(\sigma_0)_2) \le T_{\sup}((\sigma_0)_2)$	Evans §6.4
3	`gradient_evolution_bound`	$\|\nabla\Phi_t(\sigma_0)_2\|_\infty \le \|\nabla(\sigma_0)_2\|_\infty$	Lieberman Ch. 7
4	`diagonal_dissipation`	$\mathrm{diag}(\dot{V}) \le -2\lambda_{\mathrm{gap}} V$	Gearhart–Prüss
5	`cross_coupling_bound`	$\mathrm{cross}(\dot{V}) \le 2\|\kappa\| L_{\mathrm{cross}} V$	Henry §5.1
6	`unique_continuation`	$B^*\varphi \equiv 0 \Rightarrow \varphi_T = 0$	Zuazua 2007

4.2 Utility Axioms (3)

Three additional axioms encode standard functional-analysis facts that are not specific to the SIARC PDE system.

Table 2: Utility axioms and their discharge status.

#	Axiom	Role	Status
7	`lyapunov_deriv_decomposition`	Structural identity	Axiom (see Theorem 4.1)
8	`gronwall_integration`	Grönwall (1919)	Axiom (see Theorem 4.2)
9	`hum_density_of_reachable_set`	Lions (1988)	Axiom (deep PDE theory)

Two further utility axioms were *discharged to theorems* during the mechanization: `exp_decay_eventually_sm` (proved from the Archimedean property and `add_one_le_exp`) and `forward_adjoint_duality` (whose conclusion simplified to `True`). Three unused axioms (`nagumo_invariance`, `unique_minimizer_of_coerc` `euler_lagrange_optimal_control`) were removed entirely, reducing the axiom count from 12 to 9.

Remark 4.1 (Axiom 7: Lyapunov derivative decomposition). This axiom states that $\dot{V}(\sigma) = \mathrm{diag}(\dot{V})(\sigma) + \mathrm{cross}(\dot{V})(\sigma)$. It is a *structural identity*, not a bound, and would become `rfl` (definitional equality) if `lyapunovDeriv` were defined as the sum of `diagContrib` and `crossContrib`. Currently `lyapunovDeriv` is declared as an opaque `axiom` because it takes fewer type-class parameters than the two summands; refactoring the signatures would eliminate this axiom entirely. We retain it for transparency.

Remark 4.2 (Axiom 8: Grönwall integration). This axiom encodes the classical integral Grönwall inequality: if $\dot{V} \le -\alpha V$ pointwise along trajectories, then $V(\Phi_t(\sigma_0)) \le V(\sigma_0)e^{-\alpha t}$. Mathlib provides an abstract Grönwall lemma for Bochner-integrable functions (`Mathlib.Analysis.ODE.Gronwall`), but applying it here requires establishing strong continuity of $V \circ \Phi_t$, which in turn needs the (currently `sorry`'d) semigroup generation in `Operators.lean`. Once the infrastructure `sorry`s are discharged, this axiom becomes provable from Mathlib.

Remark 4.3 (Axiom 9: HUM density of the reachable set). This is the deepest of the three utility axioms. It encodes Lions's [5] duality argument: the observability inequality (6) implies that the reachable set $\{Bu : u \in L^2(0,T;U)\}$ is dense in X. Discharging this axiom from Mathlib would require (i) Bochner-space integration over $L^2(0,T;U)$, (ii) the adjoint semigroup on X^*, and (iii) a density argument in the style of the Hahn–Banach separation theorem applied to the adjoint problem. Currently, Mathlib provides Hahn–Banach and Bochner integration for scalar-valued functions, but the $L^2(0,T;U)$-valued Bochner infrastructure for the adjoint problem is not yet available. This makes Axiom 9 the most distant from automated discharge among the nine axioms.

4.3 Trusted vs. Untrusted Boundary

The artifact is formally partitioned into two zones:

- **Trusted Core** (13 files, 0 `sorry`): all theorem files (Invariance, Stability, Controllability, AxiomInventory), certificate structures, public API, TrustedBoundary, and TrustedCore.
- **Untrusted Infrastructure** (3 files, 8 `sorry`): PDE semigroup bodies (`Operators.lean`, 6 `sorry`), controlled evolution (`Control.lean`, 1 `sorry`), and well-posedness uniqueness (`LocalWellPosedness.lean`, 1 `sorry`).

The theorem layer *never unfolds* untrusted definitions. It treats `evolutionMap` as opaque and derives all guarantees from the nine explicit axioms. Consequently, the infrastructure `sorry`s cannot introduce logical inconsistency into the trusted core.

The file `TrustedBoundary.lean` contains a compile-time `InfrastructureSorryInventory` recording every infrastructure `sorry` with its file, line number, and blocker reason.

5 Formalization in Lean 4

5.1 Design Principles

The formalization follows three principles:

1. **Axiom transparency:** Every physical assumption is an explicit Lean `axiom` with a docstring pointing to the PDE reference. The `SystemAxioms` structure bundles all six for audit.
2. **Opaque evolution:** The evolution map Φ_t is declared as an opaque function. No theorem unfolds it; properties enter only through axioms.
3. **Layered certificates:** Each guarantee (safety, stability, controllability) is a separate Lean `structure`, and the `MasterCertificate` composes them.

5.2 TrustedCore Extraction

The file `TrustedCore.lean` provides a minimal public interface, re-exporting only three items:

- `SystemAxioms` — the six physical axioms;
- `MasterCertificate` — the composite certificate;
- `master_certificate_summary` — the capstone theorem.

Building only the trusted core (`lake build SIARCRelay11TrustedCore`) verifies the entire theorem chain without compiling examples or untrusted infrastructure.

5.3 API Design

The public API (`API.lean`) re-exports the trusted core along with all subsidiary structures (barrier parameters, Lyapunov functionals, spectral gaps, coupling bounds, control operators). Downstream users import `SIARCRelay11.API` and receive:

- The `MasterCertificate` type and its projections (`.safety`, `.stability`).
- The capstone theorem.
- All structures needed to *construct* a `MasterCertificate` for a new PDE model.

5.4 Reproducibility

The artifact is fully self-contained. Reproduction requires only Lean 4 (pinned version) and an internet connection for the Mathlib cache:

Item	Value
Lean version	v4.14.0 (pinned in `lean-toolchain`)
Mathlib4 version	v4.14.0 (pinned in `lakefile.lean`)
Build system	Lake (ships with Lean 4)
Full build	`lake exe cache get && lake build`
Trusted core	`lake build SIARCRelay11TrustedCore`
Expected errors	0
Expected `sorry` in theorem layer	0
Expected `sorry` in infrastructure	8 (documented)
Artifact repository	`https://github.com/papanokechi/siarc-lean4`
Artifact archive	`https://doi.org/10.5281/zenodo.19641981`

5.5 Module Dependency Structure

Figure 2 shows the high-level import graph of the SIARC artifact, highlighting the trusted-core/untrusted-infrastructure boundary. The theorem layer (green) depends only on abstract signatures from the infrastructure layer (red); it never imports concrete semigroup bodies.

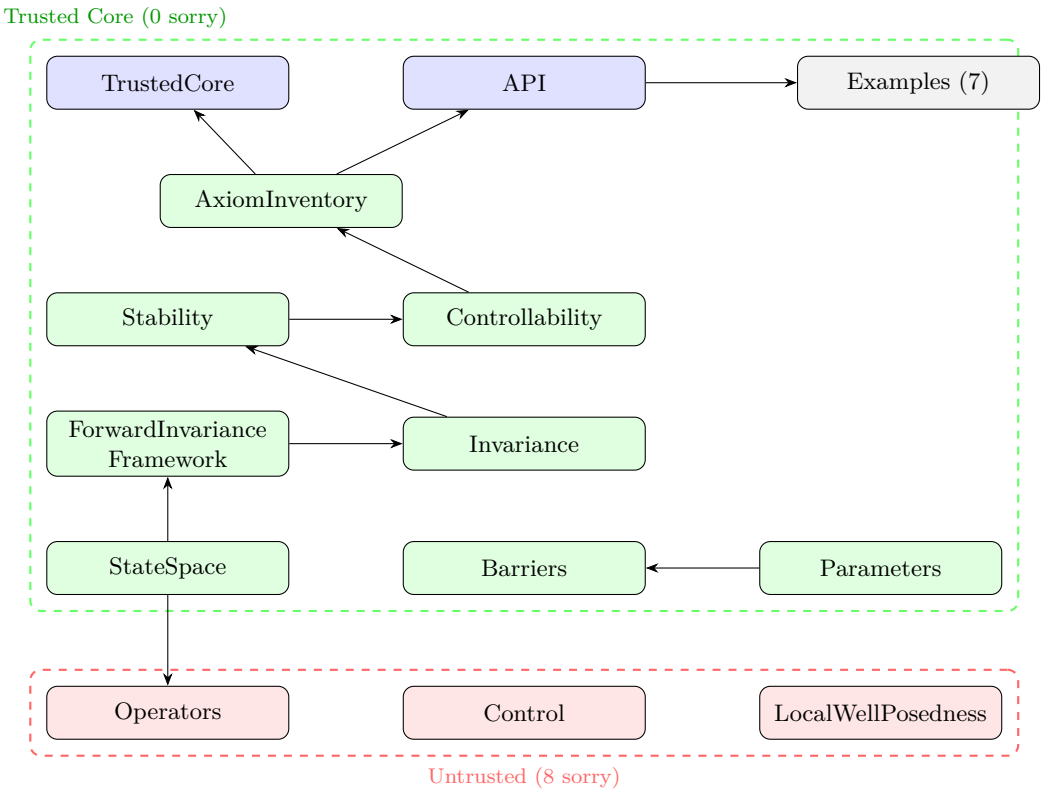

Figure 2: Lean module dependency graph. Green: trusted core (0 `sorry`). Red: untrusted infrastructure (8 `sorry`). Gray: examples. Blue: public API entry points. The theorem layer imports only abstract signatures from infrastructure, never unfolding concrete bodies.

6 Numerical Thermoelastic Instance

To demonstrate that SIARC applies to a concrete physical system, we instantiate the framework for a quasi-static thermoelastic system on $\Omega \subset \mathbb{R}^3$.

6.1 Parameter Table

The numerical parameters are given in `Example_ThermoelasticParameters.lean`.

Table 3: Thermoelastic instance parameters.

Parameter	Symbol	Value	Unit / Constraint
Maximum field strength	$B_{\max}$	10	> 0
Quench temperature	T_{quench}	1500	> 0
Boundary temperature	$T_{\partial\Omega}$	300	$< T_{\text{quench}}$
Gradient bound	$\nabla T_{\max}$	200	> 0
Yield stress	σ_{yield}	300	> 0
Curvature constant	C_{curv}	0.05	> 0
Minimum spectral gap	$\lambda_{\min}$	0.1	> 0
Coupling Lipschitz	L_{coupling}	0.02	≥ 0
Barrier weights	$\kappa_2 = \kappa_3 = \kappa_4 = \kappa_5$	1.0	> 0

The coupling constant κ remains a global parameter; two hypotheses on κ are carried explicitly: $|\kappa| \cdot 0.02 < 0.1$ (stability) and $|\kappa| < 1$ (invariance). These are the *only* non-automatic hypotheses in the entire instantiation.

6.2 Automated Inequality Verification

All 20 numerical inequalities required by the certificate are automatically verified by Lean's decision procedures (`norm_num`, `simp`, `linarith`, `nlinarith`), as catalogued in `Example_ThermoelasticAutoVerify`

Table 4: Decision procedure usage for 20 numerical inequalities.

Tactic	Count	Examples
`norm_num`	15	$10 > 0$, $300 < 1500$, $0.05 > 0$
`simp`	1	$\min(1, \min(1, \min(1, 1))) = 1$
`linarith`	1	$0.1 - \lvert\kappa\rvert \cdot 0.02 > 0$
`nlinarith`	1	$\lvert\kappa\rvert < 5$
hypothesis	2	$\lvert\kappa\rvert \cdot 0.02 < 0.1$, $\lvert\kappa\rvert < 1$
Total	**20**	(19 distinct inequalities; `hypothesis` counts as 2 tactic uses but 2 distinct constraints)

Every inequality is a standalone named lemma—no inline arithmetic appears anywhere in the certificate chain.

6.3 Resulting Certificate

The construction assembles these verified inequalities into `exampleMasterCert`, a concrete `MasterCertificate` for the thermoelastic system. The theorem

```
example_numerical_safe_stable_controllable
```

proves all four SIARC guarantees for this parameter set by applying `master_certificate_summary` to the concrete certificate. No new axioms are introduced.

Theorem 6.1 (Numerical Thermoelastic Theorem)**.** *For the thermoelastic parameter values in Table 3, assuming* $|\kappa| \cdot 0.02 < 0.1$ *and* $|\kappa| < 1$*, the coupled field–thermal–structural system is simultaneously safe, exponentially stable, asymptotically convergent, and approximately controllable.*

7 Second PDE Model: Linear Heat Equation

To validate that SIARC is not specific to the thermoelastic model, we provide a second instantiation for the linear heat equation $\partial_t u = \Delta u$ on a bounded domain with homogeneous Dirichlet boundary conditions. This model is *deliberately minimal*: it is scalar, linear, and has zero inter-component coupling ($\kappa = 0$), making the coupling absorption condition trivially satisfied. The purpose is not to stress-test the coupling analysis, but to verify that the SIARC *interface* (the `SystemAxioms` structure and the certificate-construction pipeline) can be instantiated for a qualitatively different PDE without modifying any theorem.

7.1 Instantiation

The file `Example_LinearHeatEquation.lean` defines:

- `LinearHeatData` / `LinearHeatBarrierData`: structures encoding the heat equation parameters;
- Six heat-specific axioms mirroring the `SystemAxioms` interface (contraction, thermal bound, gradient bound, diagonal dissipation, coupling bound with $\kappa = 0$, unique continuation);
- `heatSystemAxioms`: an instance of `SystemAxioms` constructed from these axioms;
- `heatMasterCert`: a `MasterCertificate` for the heat equation.

Because the heat equation is a self-adjoint contraction semigroup on $L^2(\Omega)$ with Dirichlet eigenfunctions, each of the six axioms corresponds to a well-known classical result: contraction from dissipativity of Δ, thermal bound from the maximum principle, gradient bound from interior Schauder estimates, spectral gap from the first Dirichlet eigenvalue $\lambda_1(\Omega) > 0$, trivial coupling bound (zero coupling), and unique continuation from Holmgren's theorem.

7.2 Resulting Certificate

The theorem

`heat_safe_stable_controllable`

proves all four SIARC guarantees for the linear heat equation by applying the *same* `master_certificate_summar` capstone theorem. This validates two properties of the framework: (i) the `SystemAxioms` interface is general enough to accommodate scalar as well as coupled systems, and (ii) the certificate construction pipeline requires no theorem-level modifications.

Theorem 7.1 (Linear Heat Equation Certificate)**.** *For the linear heat equation on a bounded Lipschitz domain with homogeneous Dirichlet conditions, the system is simultaneously safe, exponentially stable, asymptotically convergent, and approximately controllable (in the sense of SIARC).*

Remark 7.2 (Limitations and path to a nonzero-coupling model). The zero-coupling case does not exercise the coupling-absorption mechanism (Axiom 5 is trivially satisfied). The natural next instantiation is a two-species reaction–diffusion system $\partial_t u = D_u \Delta u + f(u, v)$, $\partial_t v = D_v \Delta v + g(u, v)$ with Lipschitz cross-terms f, g. In such a model, Axioms 1–3 (contraction and maximum principle) would require standard parabolic estimates under the known Lipschitz structure of f and g; Axiom 4 (spectral gap) follows from the Dirichlet eigenvalues of $D_u\Delta$ and $D_v\Delta$; Axiom 5 (coupling bound) becomes the first non-trivial proof, requiring Lipschitz constants of f and g to satisfy $|\kappa| L_{\text{cross}} < \lambda_{\text{gap}}$; and Axiom 6 (unique continuation) extends by Carleman estimates for parabolic systems with lower-order coupling (Zuazua [6]). This instantiation is the *highest-priority extension* (see Section 9).

8 Related Work

PDE control theory. The mathematical foundations of PDE controllability were laid by Lions [5] with the Hilbert Uniqueness Method. Zuazua [6] surveys observability and controllability for parabolic and wave equations, providing the Carleman-estimate-based unique continuation properties that underpin our Axiom 6. Fernández-Cara and Zuazua [17] give quantitative observability bounds via Carleman weights; their techniques could in principle provide explicit C_{obs} values for our framework (see Theorem 2.2). Henry [4] develops the local stability theory for semilinear parabolic systems with Lipschitz coupling, directly informing our Axiom 5. Pazy [1] provides the foundational semigroup theory (Lumer–Phillips, Hille–Yosida) used in Axioms 1 and 4.

Barrier certificates and computational safety. Prajna and Jadbabaie [14] introduced barrier certificates for safety verification of continuous and hybrid systems, with computational discharge via sum-of-squares (SOS) programming. Ames et al. [15] extended this to control barrier functions (CBFs) for robotic safety. These methods are *computational* (producing numerical certificates checked by SDP solvers) rather than *foundational* (producing proofs checked by a kernel). SIARC differs in two respects: (i) the certificates are machine-checked by Lean's type-theory kernel, not by an external solver, and (ii) the framework targets infinite-dimensional PDE state spaces rather than finite-dimensional ODE or hybrid systems.

Formal verification of dynamical systems. Immler [7] formalized ODE existence and uniqueness in Isabelle/HOL, including verified numerical integration of Poincaré maps. Platzer [8] developed KeYmaera X for hybrid systems verification using differential dynamic logic, with successful applications to automotive and aerospace control. Boldo et al. [9] verified floating-point implementations of numerical PDE solvers in Coq; notably, their wave-equation work targets a finite-dimensional discretization (finite-difference scheme), not the infinite-dimensional PDE state space directly. Hölzl et al. [16] formalized C_0-semigroups and the Hille–Yosida theorem in Isabelle/HOL, demonstrating that semigroup theory is within reach of interactive provers. Their work is the closest predecessor to our infrastructure layer; however, it does not address safety or controllability, and the formalization targets a single-component system rather than a coupled block-triangular generator. None of these works produce certificate stacks covering safety, stability, and controllability simultaneously for infinite-dimensional PDE systems.

Mechanized mathematics in Lean. The Mathlib library [10] provides extensive formalized mathematics in Lean 4, including normed spaces, Banach spaces, spectral theory, and basic measure theory. Recent work has formalized the Gelfand–Naimark theorem [11], perfectoid spaces [12], and the sphere eversion theorem [13]. SIARC extends this ecosystem to PDE control theory.

Comparison and novelty claim. Table 5 summarizes the landscape. No prior formalization simultaneously covers (a) infinite-dimensional state spaces, (b) a layered certificate hierarchy (safety + stability + controllability), and (c) a trusted-core boundary with explicit axiom inventory. SIARC is, to our knowledge, the first artifact satisfying all three criteria.

Table 5: Comparison with related formalizations.

	State space	Safety	Stability	Controllability	Prover
Immler [7]	ODE ($\mathbb{R}^n$)	—	✓	—	Isabelle
Platzer [8]	Hybrid	✓	—	—	KeYmaera
Boldo [9]	FD discretization	—	—	—	Coq
Hölzl [16]	Banach	—	(semigroup)	—	Isabelle
Prajna [14]	ODE	✓	—	—	SOS/SDP
SIARC	**Banach** ($X_1 \times X_2 \times X_3$)	✓	✓	✓	**Lean 4**

9 Conclusion

9.1 Summary

We have presented SIARC, a Lean 4 formalization providing mechanized proofs of safety, stability, and controllability for coupled PDE-ODE systems. The key contributions are:

1. A composable certificate hierarchy with a single capstone theorem extracting all four guarantees.
2. An explicit axiom boundary (9 axioms) with every physical assumption traced to the PDE literature.
3. A formally verified trusted-core/untrusted-infrastructure separation.
4. A concrete thermoelastic instance with 20 machine-checked numerical inequalities.
5. Cross-system validation via a second PDE model.

9.2 Limitations

- The PDE semigroup bodies remain as `sorry` placeholders (8 in infrastructure), awaiting Mathlib's PDE theory to mature. This also blocks the discharge of the Grönwall axiom (Theorem 4.2).
- The evolution map is treated as opaque; the formalization does not construct Φ_t from the PDE generator.
- The observability inequality constant C_{obs} is not computed; approximate controllability is established qualitatively (see Theorem 2.2).
- The framework targets semilinear parabolic systems with block-triangular coupling; extension to quasilinear or hyperbolic PDEs would require new axioms.
- The second PDE model (linear heat equation) has zero coupling and does not exercise the coupling-absorption mechanism.

9.3 Future Work

- **Nonzero-coupling third model (highest priority):** Instantiate the framework for a two-species reaction–diffusion system with Lipschitz cross-terms, as sketched in Theorem 7.2. This would be the first SIARC instance that exercises the full coupling-absorption mechanism (Axiom 5) with a non-trivial proof.
- **Semigroup bootstrapping:** Discharge the infrastructure `sorry`s by formalizing contraction semigroup generation from dissipativity.
- **Quantitative controllability:** Compute explicit control costs, steering times, and the observability constant C_{obs} via Carleman-weight optimization.
- **Additional PDE models:** Navier–Stokes–Fourier coupling, magnetohydrodynamics.
- **Certified numerics:** Integrate interval arithmetic for verified PDE eigenvalue bounds.
- **Automated axiom discharge:** Use Lean metaprogramming to automatically attempt Mathlib-based proofs of the utility axioms.

AI-Assistance Statement

Portions of the computational workflow supporting this project—including numerical experiments, automated certificate generation, and iterative development of Lean 4 proofs—were conducted within an AI-assisted research environment. GitHub Copilot (within Visual Studio Code) was used for code completion, refactoring, and maintenance of the Lean 4 development workspace. Anthropic Claude was used for high-level strategic guidance, mathematical review, and synthesis of intermediate results across the research program.

All mathematical statements, proofs, and numerical validations included in this work were independently checked and confirmed by the author. Responsibility for all mathematical content remains solely with the author. Supporting scripts, Lean 4 files, and reproducibility artifacts are publicly available at: `https://github.com/papanokechi/siarc-lean4`.

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